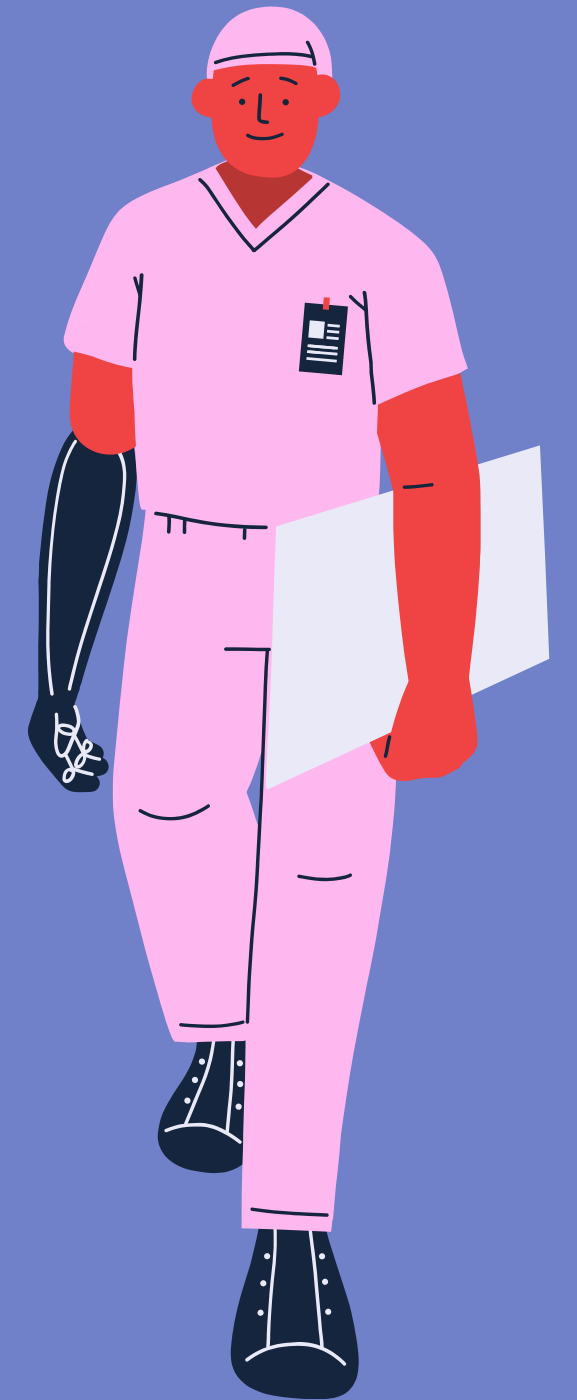


# Project Overview

**The Hospital Management Analysis project aims to analyze and optimize healthcare operations using SQL. This project provides valuable insights into patient appointments, doctor performance, prescription trends, and hospital efficiency and enhance decision-making processes, by querying a structured hospital database.**



# Table Creation:

```
-- Doctors Table
CREATE TABLE doctors (
    doctor_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    specialty VARCHAR(50),
    contact_number VARCHAR(15)
```

```
-- Prescriptions Table
CREATE TABLE prescriptions (
    prescription_id INT PRIMARY KEY,
    appointment_id INT,
    medication VARCHAR(50),
    dosage VARCHAR(20),
    instructions VARCHAR(50),
    FOREIGN KEY (appointment_id)
    REFERENCES appointments(appointment_id)
);
```

```
1 • create database Hospital_Management_db;
2 • use Hospital_Management_db;
3
4 -- Patients Table
5 • CREATE TABLE patients (
6     patient_id INT PRIMARY KEY,
7     first_name VARCHAR(50),
8     last_name VARCHAR(50),
9     gender VARCHAR(10),
10    age INT,
11    contact_number VARCHAR(15)
12    );
```

```
-- Appointments Table
CREATE TABLE appointments (
    appointment_id INT PRIMARY KEY,
    patient_id INT,
    doctor_id INT,
    appointment_date DATE,
    status VARCHAR(20),
    FOREIGN KEY (patient_id) REFERENCES patients(patient_id),
    FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id)
```

# 1) Find the average age of patients for each gender.

```
-- 1)Find the average age of patients for each gender.  
select gender,  
       avg(age) as Patients_avg_age  
from patients  
group by gender;
```

| Result Grid |        |                  | Filter Rows: |
|-------------|--------|------------------|--------------|
|             | gender | Patients_avg_age |              |
| ▶           | Male   | 48.9283          |              |
|             | Female | 49.4360          |              |



# 2)List the top 5 doctors with the highest number of completed appointments.



```
-- 2)List the top 5 doctors with the highest number of completed appointments.
select
  d.first_name as Doctors_FirstName,
  d.last_name as Doctors_LastName,
  DATE_FORMAT(appointment_date, '%Y-%m') as month,
  COUNT(*) as total_appointments
from
  appointments a
join doctors d
  on a.doctor_id = d.doctor_id
group by Doctors_FirstName,
  Doctors_LastName,
  DATE_FORMAT(appointment_date, '%Y-%m')
order by
  month;
```

| Result Grid     Filter Rows:   Export:    Wrap Cell Content:    Fetch rows: |                  |         |                    |
|-----------------------------------------------------------------------------|------------------|---------|--------------------|
| Doctors_FirstName                                                           | Doctors_LastName | month   | total_appointments |
| Dr. Michael                                                                 | Doe              | 2024-01 | 3                  |
| Dr. Sarah                                                                   | Johnson          | 2024-01 | 3                  |
| Dr. Michael                                                                 | Williams         | 2024-01 | 1                  |
| Dr. David                                                                   | Williams         | 2024-01 | 2                  |
| Dr. Emily                                                                   | Smith            | 2024-01 | 3                  |
| Dr. John                                                                    | Brown            | 2024-01 | 1                  |
| Dr. David                                                                   | Doe              | 2024-01 | 1                  |

### 3)Find the most commonly prescribed medication .

```
-- 3)Find the most commonly prescribed medication.  
select  
    medication,  
    COUNT(*) as prescription_count  
from prescriptions  
group by medication  
order by prescription_count DESC  
limit 1;
```

| Result Grid     Filter Rows: <input type="text"/> |            |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------|
|                                                                                                                                                                                                                             | medication | prescription_count |
| ▶                                                                                                                                                                                                                           | Metformin  | 326                |



# 4) Find doctors having specialty in Dermatology

```
-- 4) Find doctors having specialty in Dermatology

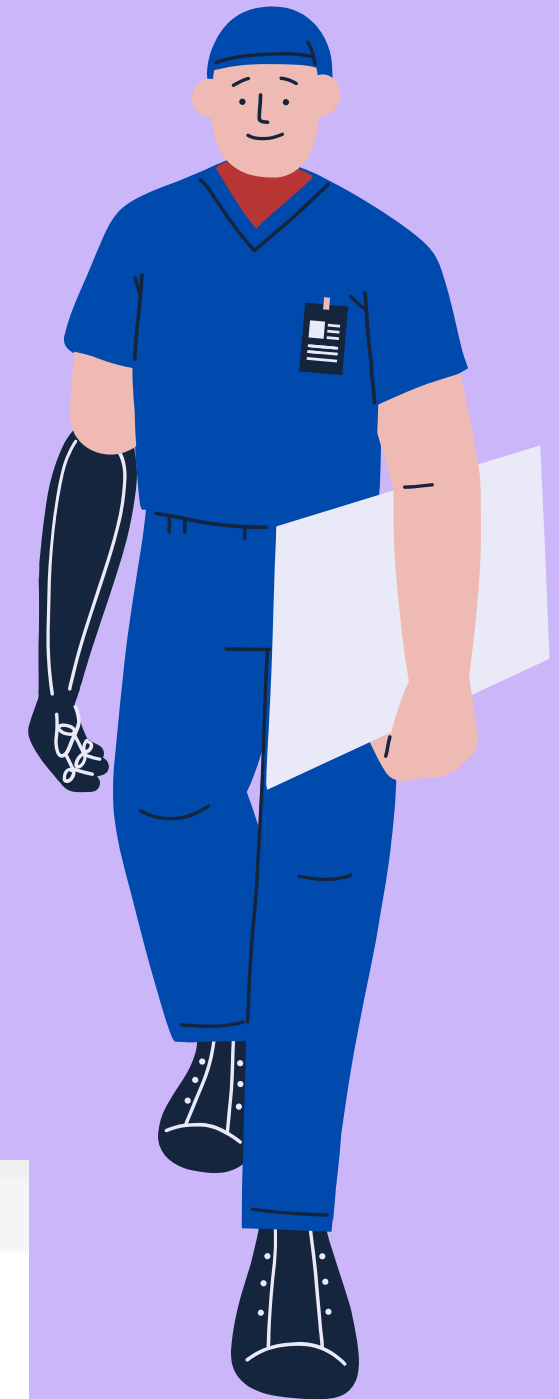
select first_name,
       last_name,
       specialty
from doctors
where specialty = 'Dermatology';
```

| id | first_name  | last_name | specialty   |
|----|-------------|-----------|-------------|
|    | Dr. Emily   | Williams  | Dermatology |
|    | Dr. David   | Williams  | Dermatology |
|    | Dr. John    | Smith     | Dermatology |
|    | Dr. Emily   | Johnson   | Dermatology |
|    | Dr. Sarah   | Smith     | Dermatology |
|    | Dr. Michael | Doe       | Dermatology |

# 5) Calculate the percentage of cancelled appointments for each doctor.

```
-- 5) Calculate the percentage of cancelled appointments for each doctor.
SELECT d.doctor_id,
       d.first_name,
       d.last_name,
       COUNT(case when a.status = 'Cancelled' then
1 else NULL end) * 100.0 / COUNT(*) as cancel_percentage
FROM doctors d
JOIN
    appointments a ON d.doctor_id = a.doctor_id
GROUP BY d.doctor_id, d.first_name, d.last_name
ORDER BY cancel_percentage DESC;
```

|   | doctor_id | first_name  | last_name | cancel_percentage |
|---|-----------|-------------|-----------|-------------------|
| ▶ | 3         | Dr. Emily   | Williams  | 55.55556          |
|   | 44        | Dr. Michael | Doe       | 44.44444          |
|   | 49        | Dr. Emily   | Johnson   | 41.30435          |
|   | 39        | Dr. Emily   | Johnson   | 41.30435          |
|   | 25        | Dr. Sarah   | Johnson   | 40.47619          |



# 6) Find the top 5 patients who have made the most appointments.

```
-- 6)Find the top 5 patients who have made the most appointments.  
select p.first_name as Patients_FirstName,  
       p.last_name as Patients_LaststName,  
       count(a.appointment_id) as Total_appointments  
from patients p  
join appointments a on p.patient_id = a.patient_id  
group by Patients_FirstName,  
         Patients_LaststName  
order by Total_appointments desc  
Limit 5 ;
```

| Patients_FirstName | Patients_LaststName | Total_appointments |
|--------------------|---------------------|--------------------|
| David              | Anderson            | 43                 |
| Bob                | Taylor              | 40                 |
| Sophia             | Garcia              | 39                 |
| Alice              | Brown               | 39                 |
| Bob                | Williams            | 37                 |

# 7) Calculate the average number of appointments completed for a doctor.

```
-- 7) Calculate the average number of appointments completed for a doctor.
select
    avg(Completed_appointments) as Avg_completed_appointments_per_doctor
from
    (select
        doctor_id,
        count(*) as Completed_appointments
    from appointments
     where status = 'Completed'
    group by doctor_id)
    as Completed_data;
```

| Result Grid |                            | Filter Rows: |
|-------------|----------------------------|--------------|
|             | Avg_completed_appointments |              |
| ▶           | 13.4400                    |              |



# 8)List the medications that were prescribed more than 5 times in a month.

```
-- 8)List the medications that were prescribed more than 5 times in a month.
select p.medication as Medication,
       date_format(a.appointment_date, '%y-%m') as Month,
       count(*) as Priscrition_Count
from prescriptions p
     Join appointments a on p.appointment_id = a.appointment_id
group by p.medication, Month
having count(*) > 5
order by Priscrition_Count desc;
```

| Result Grid    Filter Rows: <input type="text"/>   Export:  Wrap Cell Content |              |       |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|-------------------|
|                                                                                                                                                                                                                                                         | Medication   | Month | Priscrition_Count |
| ▶                                                                                                                                                                                                                                                       | Ibuprofen    | 24-08 | 10                |
|                                                                                                                                                                                                                                                         | Amoxicillin  | 25-11 | 10                |
|                                                                                                                                                                                                                                                         | Ibuprofen    | 27-10 | 10                |
|                                                                                                                                                                                                                                                         | Atorvastatin | 26-11 | 9                 |
|                                                                                                                                                                                                                                                         | Metformin    | 26-09 | 9                 |
|                                                                                                                                                                                                                                                         | Metformin    | 26-07 | 9                 |
|                                                                                                                                                                                                                                                         | Metformin    | 24-04 | 9                 |

# 9)Find the percentage of completed, canceled, and scheduled appointments for each month.

```
-- 9)Find the percentage of completed, canceled, and scheduled appointments
-- for each month.
select date_format(appointment_date, '%Y-%M') as Month,
sum(case when status = 'completed' then 1 else 0 end)*100/count(*) as completed_percent
sum(case when status = 'cancelled' then 1 else 0 end)*100/count(*) as cancelled_percent
sum(case when status = 'Scheduled' then 1 else 0 end)*100/count(*) as Scheduled_percent
  from appointments
group by Month ;
```

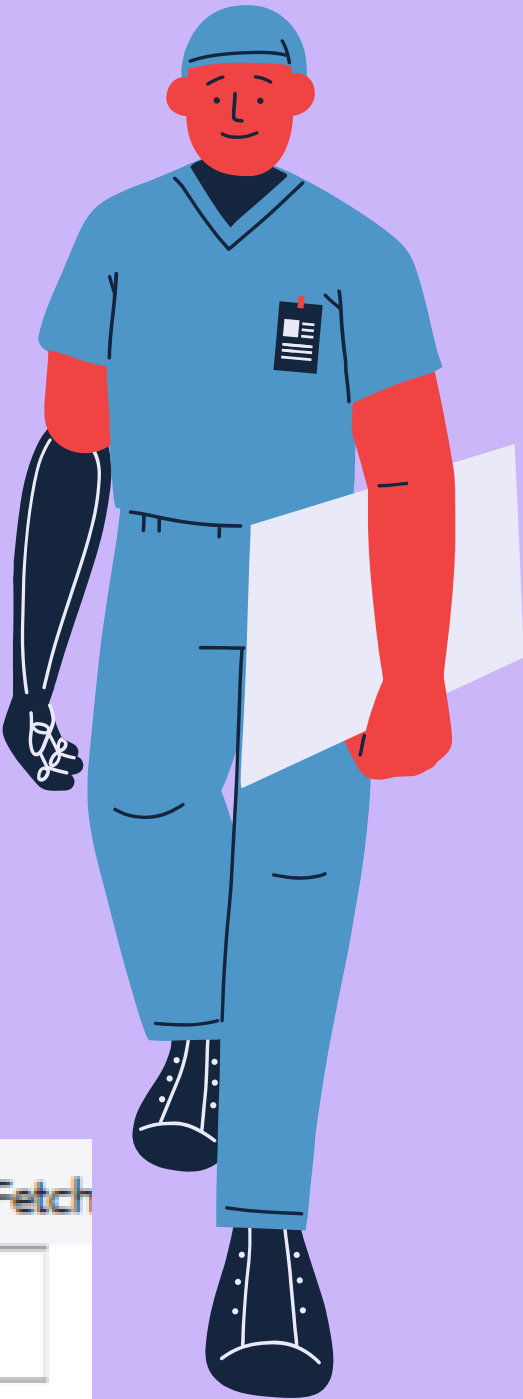
| Result Grid    Filter Rows:   Export:  Wrap Cell Content: |               |                      |                      |                      |
|-----------------------------------------------------------|---------------|----------------------|----------------------|----------------------|
|                                                           | Month         | completed_percentage | cancelled_percentage | Scheduled_percentage |
| ▶                                                         | 2024-January  | 32.2581              | 38.7097              | 29.0323              |
|                                                           | 2024-February | 34.4828              | 27.5862              | 37.9310              |
|                                                           | 2024-March    | 22.5806              | 54.8387              | 22.5806              |
|                                                           | 2024-April    | 36.6667              | 26.6667              | 36.6667              |
|                                                           | 2024-May      | 35.4839              | 38.7097              | 25.8065              |
|                                                           | 2024-June     | 20.0000              | 33.3333              | 46.6667              |



# 10)Identify which doctor has attended to the highest number of unique patients.

```
-- 10)Identify which doctor has attended to the highest number of unique patients.
select distinct d.first_name as Doctors_FirstName,
               d.last_name as Doctors_LastName,
count(*) as Unique_Patients
from appointments a
join patients p on a.patient_id = p.patient_id
join doctors d on d.doctor_id = a.doctor_id
group by d.first_name,d.last_name
order by Unique_Patients desc
limit 1;
```

| Result Grid    Filter Rows: <input type="text"/>   Export:    Wrap Cell Content:    Fetch |                   |                  |                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|-----------------|
|                                                                                                                                                                                                                                                                                                                                                          | Doctors_FirstName | Doctors_LastName | Unique_Patients |
| ▶                                                                                                                                                                                                                                                                                                                                                        | Dr. Emily         | Johnson          | 208             |



# 11)Find the number of patients who have visited the hospital more than once in the past 6 months.

```
-- 11)Find the number of patients who have visited
-- the hospital more than once in the past 6 months.
select
    p.first_name as Patients_FirstName,
    p.last_name as Patients_LastName,
    COUNT(*) as visit_count
from appointments a
join patients p on a.patient_id = p.patient_id
where
    appointment_date >= DATE_SUB(CURDATE(), INTERVAL 6 MONTH)
group by Patients_FirstName,Patients_LastName
having COUNT(*) > 1;
```

| Result Grid    Filter Rows: <input type="text"/>   Export:    Wrap Cell Content:  |                    |                   |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|-------------|
|                                                                                                                                                                                                                                                                                                                                                  | Patients_FirstName | Patients_LastName | visit_count |
| ▶                                                                                                                                                                                                                                                                                                                                                | John               | Smith             | 24          |
|                                                                                                                                                                                                                                                                                                                                                  | Sarah              | Miller            | 13          |
|                                                                                                                                                                                                                                                                                                                                                  | Alice              | Brown             | 35          |
|                                                                                                                                                                                                                                                                                                                                                  | Bob                | Johnson           | 25          |
|                                                                                                                                                                                                                                                                                                                                                  | Laura              | Taylor            | 11          |
|                                                                                                                                                                                                                                                                                                                                                  | Bob                | Taylor            | 35          |
|                                                                                                                                                                                                                                                                                                                                                  | Laura              | Garcia            | 10          |

# 12) Find the Most Recent Appointment for among the Patients Using Subquery.

```
-- 12) Find the Most Recent Appointment for among the Patients Using Subquery.
select *
  from appointments a
 where appointment_date =
    (select
       max(appointment_date)
     from appointments a
    where patient_id = a.patient_id
    );
```

| Result Grid    Filter Rows: <input type="text"/>   Edit:      Export/Import:     Wrap Cell Content |                |            |           |                  |           |
|----------------------------------------------------------------------------------------------------|----------------|------------|-----------|------------------|-----------|
|                                                                                                    | appointment_id | patient_id | doctor_id | appointment_date | status    |
| ▶                                                                                                  | 2000           | 404        | 15        | 2029-06-22       | Completed |



# 13) Calculate the Running Total of Appointments Per Month Using Window Functions.

```
-- 13) Calculate the Running Total of Appointments Per Month Using Window Functions.  
select  
    DATE_FORMAT(appointment_date, '%Y-%m') as month,  
    COUNT(*) as total_appointments,  
    SUM(COUNT(*)) over (ORDER BY DATE_FORMAT(appointment_date, '%Y-%m')) as running_total  
from  
    appointments  
group by month  
order by running_total DESC;
```

| Result Grid    Filter Rows: <input type="text"/>   Export:    Wrap Cell Co |         |                    |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------|---------------|
|                                                                                                                                                                                                                                                      | month   | total_appointments | running_total |
| ▶                                                                                                                                                                                                                                                    | 2029-06 | 22                 | 2000          |
|                                                                                                                                                                                                                                                      | 2029-05 | 31                 | 1978          |
|                                                                                                                                                                                                                                                      | 2029-04 | 30                 | 1947          |
|                                                                                                                                                                                                                                                      | 2029-03 | 31                 | 1917          |
|                                                                                                                                                                                                                                                      | 2029-02 | 28                 | 1886          |
|                                                                                                                                                                                                                                                      | 2029-01 | 31                 | 1858          |

# 14)Find the First and Second Most Prescribed Medications Using Dense Rank

```
-- 14)Find the First and Second Most Prescribed Medications Using Dense Rank

select
  medication,
  COUNT(*) as prescription_count,
  dense_rank() over (ORDER BY COUNT(*) DESC) as Rank_number
from
  prescriptions
group by
  medication
LIMIT 2;
```

| Result Grid     Filter Rows: <input type="text"/>   Export:    Wrap Cell Content:  |              |                    |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------|-------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                        | medication   | prescription_count | Rank_number |
| ▶                                                                                                                                                                                                                                                                                                                                                                                                                                      | Metformin    | 326                | 1           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                        | Atorvastatin | 302                | 2           |

# Insights

- **Appointment Trends Analysis:** Helped identify peak consultation times and patient appointment behavior
- **Doctor Performance Ranking:** Assisted in workload distribution and performance optimization.
- **Prescription Insights:** Revealed the most commonly used medications, aiding stock management.
- **Hospital Efficiency:** Showcased retention rates and patient revisit patterns for better operational planning.

# Conclusion

**The Hospital Management Analysis using SQL project helped me understand hospital operations by analyzing patient appointments, doctor performance, prescriptions, and patients behavior. I used basic to advanced SQL concepts like SELECT, WHERE, ORDER BY, and GROUP BY to retrieve and summarize data. Joins were used to combine patient, doctor, and appointment details, while aggregate functions like COUNT, SUM, and AVG helped in data analysis. I applied subqueries to fetch recent patient visits and window functions (RANK, DENSE\_RANK) to rank doctors and track trends. Time-based functions (DATE\_FORMAT, CURDATE) helped in analyzing appointment patterns. This project showed how SQL can be used to extract valuable insights for better hospital management, and it can be expanded further with data visualization and predictive analytics for even deeper insights.**